

Chapter 2: Prompt Engineering Basics

Class 8: Practical / Activities (10 Minutes)

Topic: Improve a Weak Prompt

Activity 1: Identify and Improve Weak Prompts (4 Minutes)

Objective

To help students identify weak AI prompts and improve them by adding clarity, context, audience, and output format to generate more accurate AI responses.

Materials Required

- Computer/Laptop or Tablet
 - Internet Connection
 - AI Tool (ChatGPT, Google Gemini, Microsoft Copilot, etc.)
 - Prompt Improvement Worksheet
 - Notebook & Pen
 - Teacher Observation Sheet
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Instructions

1. Open the AI tool.
2. Read the weak prompt provided by the teacher.
3. Identify what information is missing.
4. Rewrite the prompt by adding:
 - Clear topic
 - Context
 - Audience
 - Output format
 - Number of points or word limit
5. Enter both prompts into the AI tool.

6. Compare the AI responses and discuss which is better.

Sample Weak Prompts

- Tell me about pollution.
- Write about computers.
- Explain animals.
- Make a poster.
- Write a story.
- Explain science.

Example of Prompt Improvement

Weak Prompt	Improved Prompt
Tell me about pollution.	Explain the causes, effects, and prevention of air pollution in simple English using five bullet points for Class IX students .
Write about computers.	Write a 150-word article explaining the basic parts of a computer with simple examples for Class IX students .
Make a poster.	Create a poster idea for World Environment Day using green and blue colours , with the slogan " Save Nature, Save Future. "

Student Prompt Improvement Worksheet

Task	Student Response
Weak Prompt	_____
Missing Information	_____
Improved Prompt	_____
Which Response Was Better?	_____
Reason	_____

Student Activity Checklist

Students should:

- ✓ Identify weak prompts.

- ✓ Add missing details.
 - ✓ Mention the audience.
 - ✓ Specify the output format.
 - ✓ Compare AI responses.
 - ✓ Explain the improvements.
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Teacher Observation Checklist

- ✓ Identifies weak prompts correctly.
 - ✓ Improves prompts effectively.
 - ✓ Uses appropriate AI instructions.
 - ✓ Participates actively.
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Activity 2: Compare AI Responses (3 Minutes)

Objective

To help students understand how prompt quality affects the quality of AI-generated responses.

Materials Required

- AI Tool
 - Comparison Worksheet
 - Teacher Evaluation Sheet
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Instructions

1. Use both the **weak prompt** and the **improved prompt** in the AI tool.
 2. Read both AI responses carefully.
 3. Compare the responses based on:
 - Accuracy
 - Clarity
 - Detail
 - Organization
 4. Record your observations.
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AI Response Comparison Worksheet

Evaluation Criteria	Weak Prompt Response	Improved Prompt Response
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Clear Answer	☐	☐
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Detailed Information	☐	☐
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Easy to Understand	☐	☐
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Proper Format	☐	☐
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Useful for Learning	☐	☐
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Discussion Questions

1. Which response was more accurate?
 2. Which response was easier to understand?
 3. Why did the improved prompt produce a better result?
 4. How can prompt refinement save time?
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Teacher Observation Checklist

- ☒ Compares AI responses effectively.
 - ☒ Explains prompt improvements.
 - ☒ Demonstrates critical thinking.
 - ☒ Participates in discussion.
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Activity 3: Prompt Refinement Challenge (3 Minutes)

Objective

To help students independently refine prompts and create high-quality AI instructions.

Materials Required

- Prompt Challenge Worksheet
 - AI Tool
 - Teacher Observation Sheet
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Instructions

1. The teacher provides **three weak prompts**.
 2. Students improve each prompt.
 3. Generate AI responses.
 4. Choose the best refined prompt.
 5. Present the improved prompt to the class.
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Prompt Refinement Challenge

Weak Prompt	Student's Improved Prompt
Explain plants.	_____
Write about sports.	_____
Tell me about space.	_____

Student Tasks

Students should:

- ✓ Improve all prompts.
- ✓ Add context.

- ✓ Mention audience.
 - ✓ Specify format.
 - ✓ Present the best prompt.
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Teacher Observation Checklist

- ✓ Writes detailed prompts.
 - ✓ Uses prompt engineering techniques.
 - ✓ Demonstrates creativity.
 - ✓ Completes the activity independently.
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Mini Practical Assessment Rubric

Assessment Area	Marks
Identifying Weak Prompts	20
Prompt Improvement Skills	20
AI Response Comparison	20
Prompt Refinement & Creativity	20
Participation & Communication	20
Total	100 Marks

Teacher Observation Checklist

- ✓ Identifies weak prompts accurately.
- ✓ Refines prompts with clear instructions.
- ✓ Evaluates AI responses critically.
- ✓ Demonstrates effective communication with AI.
- ✓ Participates confidently throughout the activity.

Practical Time Distribution

Activity	Duration
Identify and Improve Weak Prompts	4 Minutes
Compare AI Responses	3 Minutes
Prompt Refinement Challenge	3 Minutes
Total Practical Time	10 Minutes

Expected Learning Outcomes

After completing these practical activities, students will be able to:

- Identify weak or incomplete AI prompts and recognize the information they are missing.
- Improve prompts by adding **clear instructions, context, audience, output format, and required details**.
- Compare AI-generated responses to understand how prompt quality affects output quality.
- Apply prompt refinement techniques to produce accurate, organized, and meaningful AI responses.
- Develop critical thinking and effective communication skills while interacting with AI tools.
- Demonstrate responsible and productive use of prompt engineering for learning, research, and classroom activities.